

Web Technology (BCS502)

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to:		
CO 1	Understand the fundamental concepts of web development, including the history, protocols, and tools. Apply HTML and XML in the development of web projects.	K ₃ , K ₆
CO 2	Apply CSS for designing and styling web pages, including the use of CSS properties, styling elements, and advanced techniques for creating responsive web sites.	K ₂ , K ₃
CO 3	Develop interactive web applications using JavaScript and AJAX, with a focus on scripting documents, forms, and networking concepts such as internet addressing and TCP/IP sockets.	K ₃ , K ₆
CO 4	Design and implement server-side applications using Enterprise Java Beans (EJB) and Node.js, including the creation of JavaBeans, RESTful APIs, and database operations with MongoDB.	K ₂ , K ₄ , K ₆
CO 5	Implement web server functionality using Servlets and Java Server Pages (JSP), focusing on handling HTTP requests, session tracking, and utilizing custom tag libraries for dynamic web content.	K ₂ , K ₃ , K ₄
DETAILED SYLLABUS		3-0-0

Unit	Topic	Proposed Lecture
I	Introduction: Introduction and Web Development Strategies, History of Web and Internet, Protocols Governing Web, Writing Web Projects, Connecting to Internet, Introduction to Internet services and tools, Introduction to client-server computing. Web Page Designing: HTML: List, Table, Images, Frames, forms, XML: Document type definition (DTD), XML schemes, Object Models, presenting and using XML, Using XML Processors: DOM and SAX.	08
II	CSS: Creating Style Sheet, CSS Properties, CSS Styling (Background, Text Format, Controlling Fonts), Working with block elements and objects, Working with Lists and Tables, CSS Id and Class, Box Model (Introduction, Border properties, Padding Properties, Margin properties) CSS Advanced (Grouping, Dimension, Display, Positioning, Floating, Align, Pseudo class, Navigation Bar, Image Sprites, Attribute selector), CSS Color, Creating page Layout and Site Designs.	08
III	Scripting: Java script: Introduction, documents, forms, statements, functions, objects, introduction to AJAX. Networking: Internet Addressing, InetAddress, Factory Methods, Instance Methods, TCP/IP Client Sockets, URL, URL Connection, TCP/IP Server Sockets, Datagram.	08
IV	Enterprise Java Bean: Creating a JavaBeans, JavaBeans Properties, Types of beans, Stateful Session bean, Stateless Session bean, Entity bean. Node.js: Introduction, Environment Setup, REPL Terminal, NPM (Node Package Manager) Callbacks Concept, Events, Packaging, Express Framework, Restful API. Node.js with MongoDB: MongoDB Create Database, Create Collection, Insert, delete, update, join, sort, query.	08
V	Servlets: Servlet Overview and Architecture, Interface Servlet and the Servlet Life Cycle, Handling HTTP get Requests, Handling HTTP post Requests, Redirecting Requests to Other Resources, Session Tracking, Cookies, Session Tracking with Http Session Java Server Pages (JSP): Introduction, Java Server Pages Overview, A First Java Server Page Example, Implicit Objects, Scripting, Standard Actions, Directives, Custom Tag Libraries	08